Beating the Index

Item-Number Enumeration, Anonymous Market Comps, and the Economics of Sub-Minute RAM Arbitrage on eBay

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ABSTRACT

Resale arbitrage on eBay depends on seeing an underpriced listing before other buyers and knowing what it will really sell for. For used computer memory, both are harder than they look. eBay's search index surfaces a new listing a median of 256 s after it starts, and never sooner than 137 s in our data. Sold-price data sits behind a signed-in research tool whose sessions expire unpredictably. We describe and measure a two-part system run in production for two weeks. RAM Live finds new listings by walking eBay's item-number counters directly, and `ram_platform` values each one against exact-part-number sold comparables from Terapeak, reached through an anonymous endpoint we show to be equivalent to the signed-in tool. Across 24,549 RAM listings detected over 5.5 days, enumeration read listings a median of 5.1 s after they started, about 50× faster than search and within roughly two seconds of the earliest moment eBay serves them. It saw a listing first in 98.7% of cases observed by both channels, and 67.7% of the RAM it found never reached any search feed. Once listings are priced honestly, however, the market is close to efficient: only 1.9% of estimated listings show positive net profit after fees. Estimated ROI strongly predicts whether a listing sells within 24 hours, rising from 1.1% to 66%, which validates the valuation model against market behavior. We also document four edge-level blocking rules deployed within 48 hours, each aimed at the single request signature carrying our traffic, and a catalog of engineering failure modes that were repeatedly mistaken for external causes.

KEYWORDS

online marketplaces, arbitrage, search index latency, identifier enumeration, price estimation, web measurement, anti-bot systems

1 INTRODUCTION

Used DDR4 and DDR5 memory is a natural arbitrage good. Modules are small, cheap to ship, and fungible by part number: a Samsung `M393A4K40DB3-CWE` is the same object regardless of who sells it, and the platform's own sales history says what it clears at. Supply arrives unevenly from server decommissions and laptop teardowns, often from sellers who do not know what they hold. In principle the strategy is simple: find listings priced below what the exact part sells for, buy, and relist.

In practice every step leaks. Search results arrive minutes late. Titles omit or garble part numbers. Lot photos show dozens of stickers at odd angles. Shipping is priced to the viewer's country. The only reliable sold data lives in a logged-in tool. And the platform actively resists automated reading. This paper reports what happened when each leak was measured and, where possible, closed.

We make the following contributions:

- A measurement of eBay's search latency for new listings (median 256 s, floor 137 s) and a demonstration that enumerating item numbers beats it by about 50× (median 5.1 s), near the platform's own publish floor (§5).
- Evidence that most RAM reachable by enumeration is invisible to keyword and category search (16,526 of 24,412 listings) (§5).
- An anonymous route to Terapeak sold data through eBay's public Sell page, matching the signed-in tool row for row (§7).
- A calibration result showing that estimated ROI orders listings by their probability of selling within a day (§8).
- A timeline of four edge blocking rules introduced in 48 hours, with the lesson that volume concentrated on one request signature invites a targeted block (§9).
- A catalog of failure modes with measured costs (§10).

2 BACKGROUND

Listing visibility. eBay assigns item numbers from a few dozen independent, monotonically rising counters. We call a counter a *series* and identify it by the first four digits of the item number. About 24–27 series are active at any time, together issuing roughly 700–1,000 numbers per second; in a fully walked series about 65% of numbers become live listings. A listing's JSON endpoint, `/itemmodules/<id>`, begins answering 2.3–13.3 s after the listing's `startDate` (median 3.2 s, p99 8.9 s, $n = 2{,}140$). Search is a separate index that ingests listings later.

Sold data. Public sold-listing search has required sign-in since July 2026. eBay's Marketplace Insights API is closed to new applicants [5], and the legacy `findCompletedItems` call was retired in 2025. What remains is Terapeak (Product Research) [6], which reports 90 days or more of sold listings, including the true accepted Best Offer price. Prior economics work on eBay has relied on internal data access for the same reason [2, 8, 9]. A design rule of this project, never relaxed, is that comparables come only from Terapeak and never from scraping sold pages.

Unit economics. For a module bought at price p with inbound shipping s and resold at R, we use

$$\text{net} = R - 0.136\,(1.09\,R) - 0.40 - 7.00 - 1.08\,(p + s), \qquad (1)$$

where 13.6% is the final value fee assessed on the sale price plus 9% buyer sales tax, \$0.40 is the per-order fee, \$7.00 is the outbound label, and 8% is buy-side tax. A sale retains about 85 cents per dollar before a \$7.40 fixed cost, so a stick must resell above roughly \$19

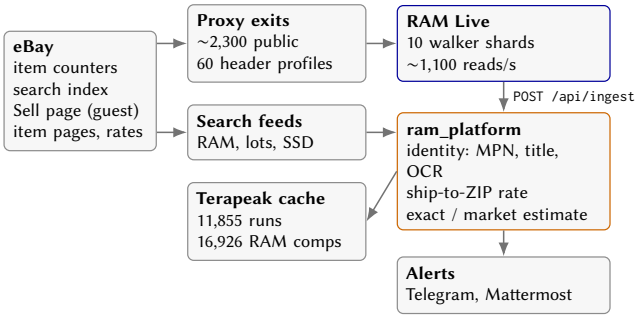


Figure 1: System architecture. RAM Live enumerates item numbers and hands every RAM hit to `ram_platform`, which also runs its own search feeds. All eBay traffic leaves through proxy exits.

to break even. Kits and lots under five sticks are priced as one sale rather than split, because splitting multiplies fixed costs.

3 SYSTEM

Both components run in a single Linux container (28 cores, 24 GB). Figure 1 shows the data flow.

3.1 RAM Live: enumeration

RAM Live runs ten walker shards and a page process. Each shard owns a load-balanced slice of the active series. For each series it predicts the newest issued number (an anchor plus rate times elapsed time, with the rate fitted to the envelope of frontier advances; pairwise fits diverge because series contain parallel sub-counters) and sweeps contiguously about one second behind that edge. Each read is one `/itemmodules` request of about 12 KB on the wire for a live listing and 1 KB for an unused number. A string-search quick parse classifies it in 0.18 ms, and only RAM titles receive a full parse. Because listings are published out of order, an unused number is settled only after a higher number has been live for 15 s; with a 1 s settle, about 11% of RAM was lost. Walker state (frontier and owed ranges) is persisted every 5 s, so a restart resumes at the lowest unread number.

At the time of writing the page process reported 1,105 reads per second across 2,310 exits on 1,593 hosts. The exit harvester had tested 5,276,941 public proxy candidates: 24,089 opened a tunnel and 2,463 (0.047%) could read eBay item data.

3.2 ram_platform: identity, comparables, alerts

`ram_platform` has tracked 151,270 listings since 16 September with 11.36 million observations. For each listing it resolves an identity (generation, form factor, ECC, speed, module size, kit layout, and ideally an exact part number) from, in order, the seller's declared MPN, the title, and OCR of the photos. It fetches a ship-to-ZIP rate and prices the listing against Terapeak comparables. An *exact* estimate requires the same part and kit layout with at least four comps and uses the units-weighted median. A *market* estimate uses the listing's specification segment, requires at least 20 comps, and is judged on the 25th-percentile resale. Priced listings are sent to Telegram and Mattermost, and every listing flagged as promising receives a slower exact re-check. A separate lot lane reads group photos of mixed lots and counts sticks by label. The Terapeak cache holds 11,855 research runs, 16,926 RAM comps and 15,912 SSD comps.

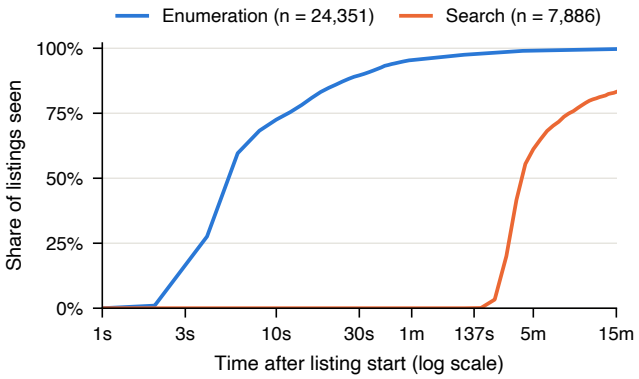


Figure 2: CDF of time from listing start to first observation. Enumeration covers all RAM hits with a valid start time; search covers hits observed by any search feed, including one polled every 2 s. The search curve reaches 83% at 900 s; the rest arrived later (p90 2,513 s).

Table 1: Latency percentiles from listing start to first observation.

Channel	min	p10	p25	p50	p75	p90
Enumeration	<2 s	3.1 s	3.9 s	5.1 s	11.7 s	31.4 s
Search	137 s	196 s	217 s	256 s	484 s	2,513 s

4 DATA AND METHODS

The primary dataset is RAM Live's hit table, copied read-only on 29 September 2026 at 21:44 UTC. It covers 24,549 RAM listings first read between 24 September 07:57 and 29 September 21:44 UTC. For each hit we have eBay's `startDate`, the time of our first live read, price and format, the time a later status check first found the listing ended, whether that ending was a sale, and the valuation `ram_platform` returned.

Detection latency is first live read minus `startDate`. *Search latency* is the earliest `ram_platform` observation from any search feed minus `startDate`. We exclude RAM Live's own hand-off, which contaminates the stored search timestamp. *Time to sale* is the first status check finding the listing sold, minus `startDate`. Checks run at 1, 3, 6, 15 and 60 minutes and less often afterward, so time to sale is an upper bound with minute-scale resolution. Other evidence comes from controlled experiments run during development (A/B tests, replays of stored responses, probes through the exit pool); their sample sizes are given where cited. Profit figures are estimates from comps unless a realized sale is stated.

5 DISCOVERY LATENCY

Figure 2 and Table 1 compare the two channels.

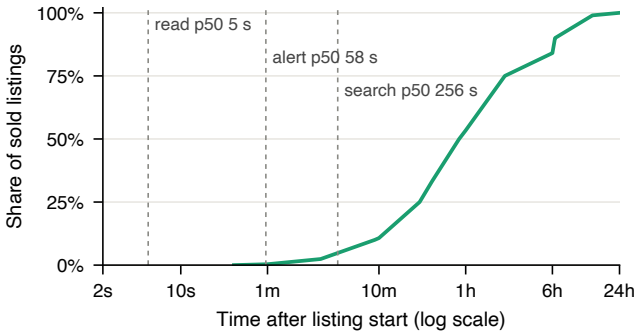


Figure 3: CDF of time from listing start to sale for the 1,987 RAM listings that sold after detection. Dashed lines mark median enumeration read, phone alert and search appearance. Sale times are upper bounds set by the status-check schedule.

Search has a hard floor. No RAM listing reached a search feed sooner than 137 s, and most arrived 180–270 s after start, independent of polling rate. An earlier study on 13,700 listings found the same floor (210–240 s) with one exception: listings scheduled to start exactly on the hour appear almost immediately, presumably because they are indexed in advance. Polling faster cannot beat an index that has not been updated.

Enumeration is near the physical floor. The median of 5.1 s is about two seconds behind the moment eBay begins serving the listing (median 3.2 s). Search observed a listing first in only 101 of 7,886 shared cases (1.3%), all during windows when the walker was lagging, such as after a restart or when a series was shed by an overloaded shard.

Most RAM is invisible to search. Of 24,412 hits with a start time, 16,526 (67.7%) were never observed by any of the five RAM search feeds. The largest single reason is geography: 10,157 hits (42%) did not ship to the US ZIP code used for pricing, and US-facing search does not show them. Commercial alerting tools we surveyed (uBuyFirst, GAlerts, Flipify, Swoopa and similar) poll search or the Browse API [4], reported at 10–15 minutes behind; none enumerates identifiers, so all inherit the index floor.

6 THE BUYER RACE

Seeing a listing first matters only if other buyers are fast. Figure 3 shows when the 1,987 detected RAM listings that later sold actually sold: 7 within one minute, 48 within three, 213 within ten, and a median of 52 minutes. The race is concentrated on the few listings that are actually cheap. An independent five-day audit using eBay's purchase history found that none of 1,665 RAM sales preceded RAM Live's first read. The fastest competing buyer took at least 38 s, and 24 listings sold within two minutes. Some early buyers were clearly faster than search permits: one SO-DIMM read at +2.3 s sold at +31 s, and was never seen by a search feed polled every 2 s.

The weak link is therefore everything after detection. The median phone alert arrived about 58 s after listing start. Of 41 listings that sold within three minutes, 16 were alerted after the sale or not at all. In one reconstructed miss, a lot listed at 17:29:04 and bought at 17:32:05 was never read. Its series had been shed by an overloaded shard at 17:20 and returned to the same shard's 10-minute cooldown. Sending shed series to a different shard with a 2-minute cooldown fixed that path.

Table 2: Guest Terapeak versus signed-in Seller Hub.

Part number	Result
`M393A4K40DB3-CWE`	Identical: 7 rows, $845.53 avg, 28 units
`HMA81GS6AFR8N-UH`	936 of 936 listing IDs match
`CMK32GX4M2D3600C18`	$148.53 vs. $148.57 avg; same $15–$300 range; 76.44% sell-through both

Table 3: Rows / exact comps per Terapeak query, before and after the part-number-only fix.

Part number	PN+MHz+GB	PN only
`HMA82GU6DJR8N-WM`	0 / 0	14 / 6
`M393A2G40DB0-CPB0Q`	4 / 2	55 / 18
`MTA16ATF2G64HZ-2G3E1`	8 / 4	109 / 41
`HMAA4GS6CJR8N-XN`	11 / 9	123 / 72

7 VALUATION

7.1 An anonymous route to Terapeak

The signed-in Seller Hub endpoint works over plain HTTP, but only inside a fresh session presenting the exact reduced browser User-Agent and matching client hints. Sessions were fragile. One lasted 19.8 hours and 1,086 lookups, while others died within minutes. Sessions held at 45–70 lookups per hour and were challenged at 136–227 per hour. Cloning a personal browser session repeatedly forced that main session to re-authenticate.

The durable route turned out to be public. eBay's logged-out Sell page includes a valuation dialog that posts a registered GraphQL query to `/sl/gql` with a guest CSRF token and receives the same Terapeak rows. We verified parity three ways (Table 2). The guest endpoint counts sold listings where Seller Hub counts units (1,662 vs. 2,648 on one query), so aggregates are rebuilt from rows. It also exposes the accepted Best Offer price: a listing at €29.99 (US $34.16) resolved to its true $28.44 sale.

7.2 Query bugs that looked like a thin market

Terapeak's `dayRange` parameter alone silently returns 30 days rather than 90; with explicit start and end dates, one part went from 1,086 to 2,490 sales. A larger bug: research queried `"PN MHz GB"`, and Terapeak conjoins every token, while sellers write "PC4-2933Y", "DDR4-2400" or "16 GB". 71% of 3,000 lookups returned zero rows, and 240 of 415 verification checks ended with "too few exact sold comps." Querying the part number alone, and letting a sanitizer filter speed and layout, restored recall (Table 3). In the first 15 live lookups afterward, zero-row results fell from 18/20 to 8/15 and mean exact comps rose from 0.1 to 3.1.

7.3 Robust resale

Terapeak rows are per listing, and one listing can represent dozens of sales, so three adjustments were needed before the median was meaningful. First, *shipping-inflated comps*: a $30 stick with $108 postage, weighted by its 21 sales, moved one baseline to $83.97 against real comps of $25–30 and produced a false 83% ROI alert. Comps whose shipping exceeds both $25 and the goods price are now dropped, and buyer postage is capped at $10. Second, *listing weight*: one listing with 8 sales moved a median to $90.47; capping each listing's weight at 3 returned it to $70.00. Third, the exact resale point is the units-weighted median, while the market tier uses the 25th percentile as a conservative gate.

7.4 Identity from photos

In one 24-hour sample, 62% of listings never resolved a part number from title or item specifics, so photos matter. The label reader tiles images to about 24 px text height, pairs module stickers one to one, and snaps reads to a lexicon of about 3,500 real part numbers. OCR used RapidOCR [14]. On 18 hand-labeled photos, correct stick counts rose from 10/18 to 18/18 and named modules from 64/120 to 115/120. On a 14-listing holdout the figures rose from 8/14 to 10/14 and from 38/60 to 56/60. The largest single gain came from not downscaling: a 960 px detector erased every label in a group photo that 1,600 px read completely. Most remaining lot errors were counting rather than recognition errors. Die-revision suffixes glued to SK hynix part numbers made one model count as two, and reads across photos could exceed the seller's stated quantity. Capping priced units at the stated count changed 90 of 3,001 lot estimates over 48 hours and removed about 15 false alerts, the largest a two-stick DDR5 listing estimated at +$795 that is really about −$84.

7.5 Sell-through without a part number

Terapeak keyword queries for a specification ("16GB DDR4 2666 SODIMM") were a poor sell-through proxy. Against 36 real parts they had a median error of 40 percentage points and gave a wrong "slow" verdict 62% of the time. Using the median sell-through of real parts with the same specification from our own cache cut leave-one-out error over 200 parts to 13 points, with 11% wrong verdicts. Across 892 hits with known sell-through, the median was 74% (IQR 57–84%).

8 MARKET EFFICIENCY

With honest comparables, very little used RAM is underpriced (Figure 4). Among 8,934 listings with any estimate, median estimated net was −$59 and only 170 (1.9%) were positive. Among 1,147 with exact comps, the median was −$37 and 20 (1.7%) were positive. Sellers asked a median of 1.23× the recent sold price. This is expected in a liquid, part-numbered market [10]: asking prices anchor high, and listings priced at the clearing price are exactly those that do not stay up.

Figure 5 is the closest available ground truth for the valuation model. The model never observes whether a listing sells, yet its ROI estimate orders listings by sale probability across two orders of magnitude, from 1.1% below −88% ROI to 66% near break-even. The plateau above zero is also informative: 35–40% of apparently

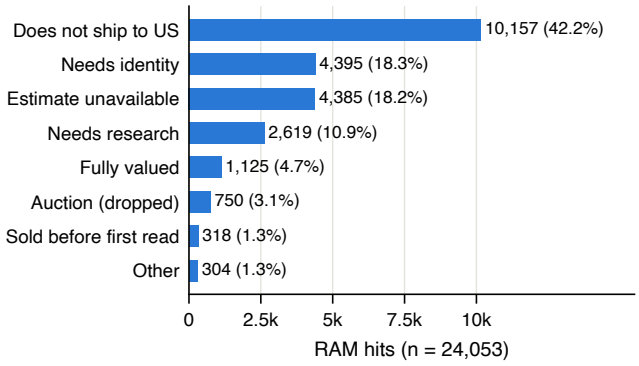


Figure 4: Final valuation status of 24,053 RAM hits. "Estimate unavailable" means no estimate could be formed (e.g., a part number conflicting with the stated kit layout), not that the item was gone.

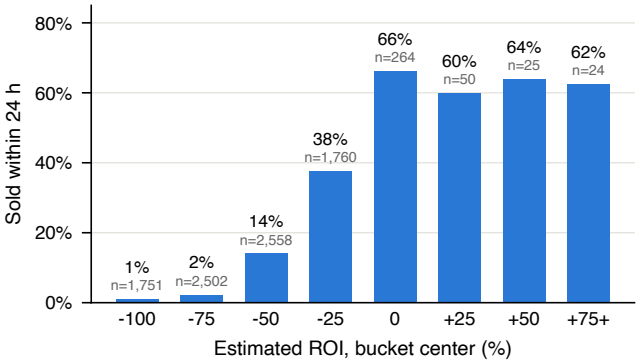


Figure 5: Share of RAM listings sold within 24 hours by estimated ROI (exact ROI where available), in 25-point buckets labeled by center; *n* is listings per bucket. The rate climbs steeply to break-even and plateaus near 60–66%.

profitable listings did not sell within a day, consistent with a mix of estimate errors (wrong identity, untested stock, niche parts) and thin demand.

Deals. During the window, 46 hits met the deal rule (net at least $1 and ROI at least 12%, sufficient comps, US shipping, adequate sell-through). They represented $4,623 of purchases at a combined estimated net of $2,448, with a median of $35.50 per deal (IQR $14.70–$63.80). 27 of 46 (59%) sold within 24 hours, at a median of 21.8 minutes after start: 2 within 3 minutes, 8 within 10 and 15 within 30. A broader seven-day count across all feeds flagged 79 deals: $9,229 to buy, $19,543 estimated resale, and about $5,700 estimated net ($4,000 on low-end estimates). The opportunity is real but small, and for most deals the window is minutes, not seconds. A human acting on a one-minute alert can win many of them.

Realized outcomes. Realized results are limited. Nine purchases totalling $706.49 are estimated at about +$174 if all sell at 90-day medians. One completed resale, a 2×16 GB kit bought for $83.58, sold for $120 (about +$12.77 net). An audit of 25 older listings on

the seller account found none sold in 5–64 days, most priced 15–55% above Terapeak medians, so the relist side needs the same discipline as the buy side.

9 ADVERSARIAL DYNAMICS

Within roughly 48 hours, eBay's edge introduced four distinct blocking rules, each matching the single request signature carrying our volume at the time (Table 4). The pattern generalizes. At 650–1,300 reads per second from one code path, any fixed byte in the request (a TLS ClientHello [1], a header value, a query parameter) becomes a cheap blocking rule. Spreading volume across realistic variants and tracking per-variant wall rates kept the system readable. At writing, no profile's wall share exceeded 0.45. Walls remain common (28–45% per profile through free exits), mostly from IP reputation rather than request shape. The third rule could have been defeated by minting full bot-management cookies in a real browser and distributing them across exits; we treated that as detection evasion and did not build it (§12).

Proxy economics. Free public proxies are a poor substrate [13]. For search pages, about 1 in 4,000 fresh candidates passed a real search, and only 3–5 were usable at once. For item reads the yield was higher (0.047%), and one clean IP sustained 24 reads per second for 18 minutes with essentially no walls; blocks began near 256 per second per IP. Search discovery used about 37 GB of transfer per day. Full enumeration coverage would need about 700 GB per day, which rules out metered residential proxy plans and points to flat-rate static addresses.

Session warmth, not headlessness. Whether a browser ran headed or headless was not what triggered search walls; session warmth and request velocity were. The first search in a fresh session succeeded 0/12 times, but the next search in the same session after a wall succeeded 15/21. Keeping the session and exit after a wall rather than discarding both raised live search success from 0/19 to 41/52 on the same exits and queries. A stealth browser [3] was statistically indistinguishable from Chromium through the pool (16/24 vs. 13/24, Fisher's exact $p = 0.56$), consistent with the browser-fingerprinting literature's view that network and behavioral signals dominate [11].

10 FAILURE MODES

Most engineering time went to a small set of recurring bug shapes (Table 5). Each first looked like a market or network problem. In every case the fix began with a query against stored data rather than a new hypothesis about the network. A related operational lesson: raising a limit on a backed-up queue releases the whole backlog. Lifting a research limit released verification checks up to 17 hours old, which posted 26 stale "new" deals in 30 minutes. Anything user-facing now carries an age gate before its limits change.

11 DISCUSSION

The bottleneck moved three times. It began as discovery: a minutes-late index and walled search. Enumeration moved detection to the platform's publish floor. It then became valuation: signed-out research, starved queries, inflated comps and miscounted lots. The anonymous Terapeak route and the fixes in §7 addressed that. It is now the last mile: alert latency, residual identity gaps, and the decision to buy.

The efficiency result should temper expectations. A market in which 98% of honest estimates are negative has no easy money, only occasional errors, captured by whoever reads fastest and trusts their comparables. Figure 5 is the strongest argument that the comparables can be trusted; its plateau is the strongest argument that no single estimate should be trusted blindly. Several small rules had large effects. Pricing sub-five-stick kits as one sale, and refusing $/GB extrapolation to capacities never actually sold, each removed a class of phantom profit; an early sweep produced 224 fake flips, mostly sibling DDR5 kits priced against one another.

12 ETHICS AND TERMS OF SERVICE

This system reads a commercial platform at high volume through third-party proxies and adapts when blocked, which is in tension with eBay's user agreement [7]. We state this plainly. The project held several lines:

- It never scraped sold search.
- It never sent eBay traffic from the operator's home IP, which is also a seller identity.
- It never solved CAPTCHAs, automated credential entry, or operated multiple accounts.
- It declined to mint and redistribute browser bot-management cookies.

The public proxies used were not necessarily intended for this use by their operators [13]. Anyone extending this work should consider sanctioned channels first; the Browse API was evaluated and rejected here only for its reported feed lag. Buyers also bear real risk. During the window, one $431 purchase proved to be part of a ring of at least six zero-feedback accounts created within three days, each with matching bait listings. A system that surfaces the cheapest listings surfaces scams disproportionately, so seller age and feedback belong in the deal rule.

13 LIMITATIONS

The enumeration window is 5.5 days, during which the system changed several times daily, so distributions pool several versions. Sale times are upper bounds from a check schedule. Search latency covers only listings search eventually observed, which biases its tail downward. Deal profits are estimates, and the realized sample is a single sale. About 22% of RAM sat on unwalked series at audit time, and scheduled listings, numbered hours before going live, read as unused when walked. Valuation uses US-marketplace data only. Valuation status is the latest stored state, so some listings were priced after they stopped being buyable.

14 FUTURE WORK

Planned work includes:

- Send first alerts before the ship-to-ZIP quote arrives and correct them in place.
- Prefetch Terapeak for every known part number on a schedule.
- Weight comparables by recency (for example, a 30-day half-life), backtested first.

Table 4: Edge blocking rules observed against RAM Live, 25–26 September 2026 (UTC).

Onset	Rejected	Blind	Response
25 Sep 22:00	aiohttp TLS fingerprint (curl from same IP passed)	~1.5 h	Stable real-browser TLS fingerprint per exit via curl_cffi [12]
26 Sep 07:35	Exact `Accept-Language: en-US,en;q=0.9`, or none	~1 h	60 browser/header profiles; retire a profile only on evidence from ≥6 exits
26 Sep 15:31	Item reads without Referer and warmed session cookies	~4 h	One homepage warm-up per session, then item Referer
26 Sep 22:10	Any `/itemmodules` URL carrying `module_groups=X`	~20 h	Bare `/itemmodules /<id>`; the parameter was ignored anyway

Table 5: Recurring failure modes. In each case the symptom pointed at an external cause while the cause was internal.

Failure	Apparent symptom	Measured cost
Glued server HTML (“9 bidsFree delivery”) defeats \b word boundaries	Auctions leaking into lots; unknown shipping	0/1,011 auction cards caught before fix, 1,011 after; shipping known 34%→58%
Prices and shipping rendered in the exit’s currency	$179 kit shown at $3,838; “$408.88 shipping”	35/362 stored prices wrong; 22 of 40 “unspecified” listings actually shipped free
`proxy_slots` was a worker-lane count, not an exit budget	“Discovery slow; exits bad”	9,458 slot errors in 6 h; four worker types dead for 15 h
SQLite `synchronous` pragma is per connection [15]	“database is locked” storms	5,363 lock errors in 6 h; ~700/h to 0 after fix
Maintenance re-queued evaluations with a per-minute key	Slow, overloaded host	1.39M pending jobs for 73k listings; 2.15M-row table cut to 352k
Every cent of valuation drift edited the alert	Alerts hours late	Bot rate-limit bans of ~13 h and ~20 h; median queue wait 7 h
Shipping text (“Free shipping”) passed to `float()`	“Terapeak works but stores nothing”	0 of 510 research runs stored a comp
“-32 PC UDIMM” parsed as 32 pieces	+$7,955 alert on a $290 stick	19 of 60k listings affected

- Persist harvested exits across restarts; one full restart shrank the healthy pool from about 3,300 to 600.
- Add seller age and feedback to the deal rule.
- Build a realized-profit ledger joining buys, relists and sales.

15 CONCLUSION

Enumerating item numbers turns eBay discovery from a four-minute problem into a five-second one, close to the rate at which the platform can serve a listing at all. Anonymous access to Terapeak’s own rows makes exact-part valuation possible without a fragile signed-in session. With both in place, the used-RAM market is efficient in aggregate and exploitable only at the margins: about 2% of listings are priced below their honest resale value, and most of those are gone within the hour. The remaining value lies not in reading faster but in faster, better-trusted decisions about the few listings worth buying.

A PRODUCTION PARAMETERS

Final value fee	13.6% of (sale + 9% tax) + $0.40
Outbound label	$7.00 per sale
Buy-side tax	8%
Deal gate	net ≥ $1 and ROI ≥ 12%
Promising flag	ROI ≥ −25%
Sell-through floor	35% (used, 90 days)
Exact estimate	≥4 comps; units-weighted median; listing weight ≤3
Market estimate	≥20 segment comps; judged on p25
Kit split threshold	5 sticks
Guest Terapeak pacing	≤600/h, 3 workers
Walker settle	15 s; reprobes at 1.5/3/6/8 s

B DATA PROVENANCE

Enumeration data: RAM Live hit table (24,549 rows), read-only copy taken 29 September 2026, 21:44 UTC, deleted after analysis. Search timing: `ram_platform` observations table (11.36M rows), opened read-only. Terapeak figures: research cache. Experimental figures come from engineering notes of 15–29 September 2026, each measured on the production container or a sandbox copy. Proxy addresses, credentials, webhook identifiers, buyer details and order numbers are omitted.

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